Harshil Modh

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Education

Stevens Institute of Technology **USA**
Master of Science in Computer Science; GPA: 3.9/4.00 *2024 – 2026*
LDRP Institute of Technology and Research **India**
Bachelor of Engineering, Computer Science *2019 – 2023*

Experience

Software Engineer, AI **Jan 2026 – Present**
GenAI Consulting *Princeton, NJ*
- Built and deployed a full-stack analytics platform (React 19, Python/FastAPI, SQL warehouse, Docker) serving live data on every request across 71K+ rows in 10 tables, deployed to Azure Container Apps.
- Architected a FastAPI backend caching a dimension table at startup and running per-request filtered SQL over 41K TRx and 27K call records; managed-identity auth with no stored credentials and JSON-repair parsing on malformed model output.
- Integrated LLM APIs into two agents and a guarded copilot, computing all scoring in Python before the model so business logic stays deterministic and testable rather than model-generated.
- Wrote 92 hermetic CI tests (no network or DB) and a 4-scenario LLM guardrail suite; re-architected deployment to scale-to-zero, eliminating $263/mo in idle billing.

Teaching Assistant, CS 546 Web Programming **Sept 2025 – Present**
Stevens Institute of Technology *Hoboken, NJ*
- Designed and published **canvas-grade-manager**, an npm CLI on the Canvas API automating late-penalty removal, bulk submission downloads, and grading reports; 1,300+ weekly downloads, adopted by 6+ TAs across two semesters.
- Built an automated grading framework asserting HTTP responses, JSON contracts, auth, and DB state, cutting turnaround 80%.
- Debugged 50+ defects across async race conditions, middleware pipelines, and auth failures, cutting resolution time 60%.
- Mentored 100+ graduate developers, reviewing 80+ weekly submissions and cutting the critical defect rate 30%.

Software Engineer **June 2023 – June 2024**
Grownited *India*
- Architected modular CRM backends in Node.js, Express, and MongoDB with layered services and standardized API contracts.
- Owned API security for 15+ production REST endpoints with JWT auth, RBAC, and validation middleware, backed by automated tests over authorization failure paths and zero privilege-escalation incidents.
- Cut API latency 50% and DB reads 40% via query-plan analysis, aggregation and index tuning, and Redis TTL caching.
- Built role-aware React and Angular dashboards with protected navigation, multi-step forms with inline validation, and Google Maps search; containerized with Docker, shipped via GitHub Actions CI/CD to AWS, reviewed PRs, mentored juniors, and ran Agile sprints.

Projects

CodePulse, AI Code Health Platform | GitHub
- Built a GitHub-integrated platform on Node.js, Express, PostgreSQL, Prisma, and BullMQ scoring repositories 0–100 on every push, fanning out 5 parallel jobs across complexity, vulnerabilities, dead code, test coverage, and architectural drift into a weighted score in under 60s.
- Engineered an AI agent platform on the OpenAI API: 6-tool RAG chat, root-cause investigator, multi-agent design debate, and guided codebase tour; streamed over SSE at sub-400ms time-to-first-response and shipped an MCP server exposing agents to Claude and Cursor.
- Implemented semantic architectural-drift detection with OpenAI embeddings and pgvector cosine search under 50ms, clustering files via K-Means with GPT-generated module summaries across a multi-tenant schema of 14 models.
- Built real-time delivery on Redis Pub/Sub and Socket.IO for live score updates, and pushed insights into the developer workflow through automated PR health comments, embeddable SVG status badges, and weekly digest jobs.
- Developed the Next.js/TypeScript client with Clerk auth, D3 and Recharts visualizations, and Cmd+K semantic search; integrated Stripe Free/Pro/Team billing with HMAC-verified GitHub and Stripe webhooks and hashed API keys for programmatic access.

CareConnect, Real-Time Caregiving Platform | GitHub | Live
- Built a full-stack MERN app letting families and caregivers coordinate tasks, medications, vitals, and documents in one place; secured 40+ REST endpoints with 5-role access control, per-membership permission checks, and JWT plus Firebase auth, with zero cross-group data exposure.
- Cut database queries 50% with Redis caching and token-based stateless sessions; stored documents on AWS S3 with pre-signed URLs and deployed a four-service stack with Docker on AWS EC2.
- Added real-time group chat on Socket.IO with message edit history, read receipts, and admin-only delete, plus a one-tap emergency alert reaching every group member by in-app notification, chat message, and email.
- Automated scheduling with four background jobs covering recurring task generation, overdue detection, due-soon reminders with duplicate suppression, and notification cleanup; built medication dose tracking that stays accurate when multiple caregivers log concurrently.

CoMP Ledger, AI Content Attribution Platform | GitHub **AWS Hackathon Winner**
- Built a Node.js, Express, and PostgreSQL platform on AWS Bedrock proving whether an AI model's answer came from a publisher's copyrighted reporting, decomposing each answer into factual claims with Claude and scoring them against the publisher's archive.
- Designed the exclusivity scoring model measuring whether a claim appears anywhere else on the web, scoring inversely to independent source count so common knowledge scores low and original reporting scores high.
- Engineered a research pipeline checking every claim against five external sources (Wikipedia, Google, OpenAlex, GDELT, Wayback), routing evidence to a Claude judge for a structured verdict streamed over SSE across 40 endpoints; exported SHA-256 signed evidence to S3 as a tamper-evident licensing record.

Skills

Languages:	JavaScript, TypeScript, Python, Java, SQL, C, C++
Frontend:	React, Next.js, Angular, Tailwind CSS, Bootstrap, shadcn/ui, TanStack Query, Framer Motion, Recharts, D3
Backend:	Node.js, Express, FastAPI, Flask, REST APIs, GraphQL, Socket.IO, SSE, Redis, BullMQ, Prisma, Zod, Nodemailer, Cron Jobs
Databases:	PostgreSQL, MongoDB, pgvector, Qdrant, MySQL, Firestore, DynamoDB
Cloud / DevOps:	AWS (EC2, S3, Bedrock), Azure Container Apps, Docker, GitHub Actions, CI/CD, Linux, pytest, Git, Postman, Octokit, Cursor, GitHub Copilot
AI / LLM:	OpenAI API, Claude, Google Gemini API, LangGraph, LangChain, RAG, MCP, embeddings, vector search, Vapi.ai, Deepgram, Vercel AI SDK, Inngest, E2B
Auth & Billing:	Firebase Auth, Clerk, JWT, OAuth, RBAC, HMAC-SHA256, Rate Limiting, Stripe